

Quote-Locked Cross-Chain Consumer Checkout: An ERC-7683 Intent and EIP-712 Firm-Quote Architecture for Zero-Delta, Higher-Conversion Stablecoin Settlement

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Abstract

Cryptocurrency checkout flows remain hostile to ordinary consumers for two structural reasons: the fee a buyer sees at asset selection routinely differs from the amount charged at confirmation, and the buyer is forced to reason explicitly about bridges, chain identifiers, gas tokens, and which stablecoin variant they happen to hold. We argue that both failures share a single root cause—the prevailing checkout model exposes a disclaimed estimate rather than a firm, signed price—and we propose a quote-locked architecture that eliminates it. Our design combines an ERC-7683 cross-chain intent, an EIP-712 firm-quote engine with a signed lock window, a single hot-wallet solver fronting native USD Coin (USDC) inventory on Arbitrum, and Circle’s Cross-Chain Transfer Protocol (CCTP) V2 Fast Transfer for canonical, mint-and-burn normalization. The architecture disentangles pricing risk (borne by the solver inside the lock window) from settlement risk (absorbed asynchronously by the inventory pool), so the buyer is presented exactly one number that never moves against them. We evaluate the design through a deterministic, seeded A/B simulation of 1,500 orders that contrasts a re-quote-at-settlement baseline (Variant A) against the quote-locked checkout (Variant B), sweeping lock time-to-live (TTL), hedge buffer, and inventory policy. The locked variant drives the quote-vs-final delta breach rate to 0.2% with a buyer-favorable mean signed delta of -0.94 basis points (bps), lifts conversion by $+17.7\%$ relative among simulated buyers drawn from an explicit behavioral conversion model ($p \approx 3.9 \times 10^{-10}$, statistical power ≈ 1.0), achieves a p95 time-to-settlement of 15.9 s at a 1.7% fill-failure rate, and holds all-in cost to 11.5 bps against a self-funding solver subsidy of 4.2 bps below an 8.0 bps captured spread. All five pre-registered validation criteria pass. We discuss the compute-versus-quality and subsidy-versus-coverage trade-offs, the adversarial-drift stress test, scalability, and the safety implications of consumer-grade chain abstraction.

1 Introduction

The promise of stablecoin payments is that a dollar-denominated digital asset can move globally, settle in seconds, and cost a fraction of a card interchange fee. The lived experience of a consumer crypto checkout falls far short of that promise. Two defects dominate. First, quote instability: the all-in cost a buyer is shown when they choose an asset is, in almost every production flow, an estimate subject to revision at confirmation. Network fees move, automated market maker (AMM) pools shift, bridge liquidity drains, and the price that is finally charged drifts—sometimes in the buyer’s favor, often not. Second, cognitive overload: the buyer is made responsible for a stack of infrastructure decisions that have nothing to do with their purchase. Which chain is the merchant on? Do I hold the “right” USDC—native, bridged, or a wrapped variant? Do I have gas on the destination chain? Is this bridge safe? Each question is an abandonment opportunity.

These two defects are usually treated as separate engineering problems—one for the pricing team, one for the wallet team—but we contend they are the same problem viewed from two angles. Both arise because the dominant checkout abstraction surfaces a disclaimed estimate computed against live, adversarial market state, and then pushes the residual risk of that estimate onto the least-equipped party in the system: the consumer. A buyer who must mentally hedge against a moving fee, and simultaneously route their own funds across chains, is being asked to act as a market maker and a bridge operator at the moment they simply want to buy something.

The technical gap is therefore not a missing bridge or a faster rollup; those exist. The gap is the absence of a firm, cryptographically enforceable price commitment wrapped around a chain-abstracted funding experience, with both delivered atomically to a non-technical buyer. Intent-based protocols and request-for-quote (RFQ) venues have shown that a solver can commit to an output, and chain-abstraction wallets have shown that a unified balance can hide chain selection. No deployed consumer checkout, to our knowledge, composes the two into a guaranteed locked quote that settles de-

terministically to native USDC on a single destination chain while measuring its own quote-vs-final delta and abandonment.

This paper closes that gap. Our specific technical contributions are:

1. A quote-lock formulation that models the checkout as a constrained optimization over lock TTL, hedge buffer, and solver inventory, with an explicit fidelity (delta), adversarial-drift, and capital-cost objective, yielding a hard EIP-712-signed guarantee that the buyer is never charged more than the locked bound.
2. A composed architecture that joins ERC-7683 intents, an EIP-712 firm-quote engine, a single hot-wallet USDC solver on Arbitrum, and CCTP V2 Fast Transfer, disentangling pricing risk from settlement risk so the merchant payout is non-blocking and the buyer signs once.
3. A reproducible A/B evaluation harness (deterministic, seeded, $N = 1,500$) with transaction-cost-analysis (TCA) and e-commerce experiment statistics—paired bootstrap confidence intervals on delta-bps and a two-proportion test on conversion—that self-instruments the very metrics the industry leaves unmeasured.
4. An empirical demonstration, in a seeded simulation under an explicit behavioral conversion model, that locking the quote is not merely a user experience (UX) nicety but a measurable conversion driver: a +17.7% relative lift with overwhelming statistical significance, achieved at a self-funding subsidy.

2 Related Work

Intent-based and RFQ settlement. The intellectual lineage of our approach runs through intent-centric architectures, in which a user declares a desired outcome and a competitive set of solvers commits to realizing it [1]. UniswapX [2] formalizes this for swaps via Exclusive Dutch Orders, where the output range and a deadline bound the price a filler may give; on layer-2 (L2) to L2 routes it reports sub-three-second fills. Across [3] runs an optimistic intent/solver network with a bonded relay model and sub-minute cross-chain fills, separating the fast user-facing fill from slow canonical settlement—a separation we adopt and generalize. RFQ venues such as 0x [4] and Hashflow [5] demonstrate that market makers can stream firm quotes signed off-chain, eliminating AMM slippage at the cost of maker inventory risk. Our quote-lock is, in essence, a consumer-checkout-shaped RFQ: a single solver streams an EIP-712 firm quote, but unlike trading-desk RFQ we wrap it in a chain-abstracted funding flow and measure its effect on conversion, not just price improvement.

Standards: intents, signing, and settlement. ERC-7683 [6] standardizes cross-chain intents—the CrossChainOrder and ResolvedCrossChainOrder structures—so that orders are portable across solver networks; we target this interface directly. EIP-712 [7] provides typed, human-readable structured-data signing, which is the cryptographic substrate of our lock: the buyer signs a typed LockedQuote whose hash binds price, bound, and expiry. Account abstraction via ERC-4337 [8] and the ERC-20 token standard [9] underpin the gasless, single-signature experience. Circle’s CCTP [10] performs native USDC normalization by burning on the source domain and minting on the destination, avoiding the wrapped-asset and honeypot risks endemic to lock-and-mint bridges; CCTP V2 Fast Transfer compresses the attestation latency that previously made canonical USDC movement too slow for checkout.

Chain abstraction. Particle Network Universal Accounts [11] present a unified cross-chain balance and a single signature, hiding chain selection and gas entirely; Privy [12] provides embedded email/social-login wallets that remove seed-phrase friction. These systems solve the funding half of our problem but, on their own, still inherit a live-rate quote at settlement. Our contribution is to bind chain abstraction to a firm quote so that the unified balance is debited against a number that cannot move.

Bridges, aggregators, and rollups. Cross-chain liquidity aggregators—LI.FI [13], Squid [14], and Rango [15]—span 30–85+ chains and excel at routing, but their breadth comes at the price of disclaimed, route-dependent estimates rather than atomic guarantees. Systematizations of bridge security [16] catalogue the catastrophic failure modes (lock-and-mint exploits, validator compromise) that motivate our preference for canonical CCTP settlement on a single, mature destination rollup. Arbitrum [17] and the broader optimistic-rollup design space [18] provide that destination: cheap, fast, and deep in native USDC liquidity. Consumer on-ramps and payment processors—Binance Pay, Coinbase Commerce, MoonPay, and Transak—anchor our baseline matrix; they typically expose estimate-grade pricing with variable fill times.

Adversarial robustness and MEV. The hedge-buffer component of our objective is, in spirit, an adversarial-training problem: the solver must price robustly against worst-case price drift over the lock window, just as a robust model is trained against worst-case perturbations. The maximal-extractable-value (MEV) literature [19] characterizes the adversarial environment in which any on-chain commitment lives, and AMM microstructure [20] characterizes the slippage that firm quotes are designed to eliminate. We treat adversar-

ial drift as a first-class term rather than an externality pushed onto the buyer.

Measurement methodology. Our evaluation borrows from two mature disciplines. Transaction cost analysis [21] provides the all-in-cost and implementation-shortfall framing for delta-bps. Controlled online experimentation [22] provides the A/B discipline: bucketing, power analysis, and guardrail metrics. We use the nonparametric bootstrap [23] for delta-bps confidence intervals and standard two-proportion power analysis [24] for conversion sizing. The closest conceptual analogue to our self-instrumenting funnel is the practice of treating product metrics as disentangled, independently estimable signals—here, separating the pricing-fidelity signal from the conversion signal so that one cannot silently confound the other.

In summary, prior systems each solve one facet—firm quoting (RFQ, UniswapX), fast cross-chain fills (Across), canonical settlement (CCTP), or chain abstraction (Particle, Privy)—but none composes all four into a measured consumer checkout with a hard quote guarantee. That composition is our contribution.

3 Problem Formulation

Setting. A buyer holds an asset a on a source chain $c \in \mathcal{C}$ and wishes to pay a merchant a notional V denominated in native USDC on the destination chain (Arbitrum, c^*). Let the quote engine, queried at time t_0 , return a locked quote $\text{LQ} = (q, \bar{q}, \tau, h)$, where q is the quoted source-asset amount, $\bar{q}$ is the signed upper bound the buyer can be charged, τ is the lock TTL, and $h = H(\text{LQ})$ is the EIP-712 typed-data hash. The buyer’s signature σ over h produces an ERC-7683 intent $\mathcal{I}$. Settlement occurs at $t_s \leq t_0 + \tau$ at realized price p .

Delta and breach. Define the quote-vs-final delta in basis points (bps) relative to notional,

$$\delta = 10^4 \cdot \frac{p - q}{V}. \quad (1)$$

A breach is the event that the buyer is charged beyond the signed bound,

$$\mathcal{B} = \mathbf{1}[p > \bar{q}], \quad \bar{q} = q(1 + b), \quad (2)$$

where $b \geq 0$ is the hedge buffer (in bps) the solver adds to cover drift. The hard guarantee we seek is that, conditional on settlement within the lock window, the buyer never pays more than the bound, and in expectation pays no more than quoted:

$$\Pr[\mathcal{B}] \leq \epsilon, \quad \mathbb{E}[\delta] \leq 0. \quad (3)$$

Drift model. During the lock the source-asset price follows $p(t) = q(1 + \xi(t))$, where ξ is a mean-reverting drift process with adversarial tail. Concretely, ξ is an Ornstein–Uhlenbeck mean-reverting process with mean-reversion coefficient $\kappa = 0.5 \text{ s}^{-1}$ and base volatility $\sigma_0 = 6 \text{ bps}/\sqrt{\text{s}}$, scaled by a volatility-slider multiplier $\nu \in \{1\times, 2\times, 3\times, 5\times\}$; the high-drift adversarial stress test fixes $\nu = 5\times$, i.e. an effective volatility $\sigma = \nu \sigma_0 = 30 \text{ bps}/\sqrt{\text{s}}$. The solver’s exposure is the realized drift at t_s net of the buffer; the worst-case (adversarial) drift over the window is

$$\xi_\tau^{\text{adv}} = \sup_{t \leq t_0 + \tau} \xi(t). \quad (4)$$

A larger τ widens both the buyer’s convenience and the solver’s exposure; this tension is the core trade-off.

Objective. The system chooses the policy $\theta = (\tau, b, \pi)$ —lock TTL, hedge buffer, and inventory policy π governing solver USDC reserves I —to minimize a composite objective that disentangles three risks:

$$\min_{\theta} \underbrace{\lambda_f \mathbb{E}[\delta^2]}_{\text{fidelity}} + \underbrace{\lambda_a \mathbb{E}[(\xi_\tau^{\text{adv}} - b)_+]}_{\text{adversarial drift}} + \underbrace{\lambda_c \mathbb{E}[s(\theta)]}_{\text{capital cost}}, \quad (5)$$

subject to

$$\Pr[\mathcal{B}] \leq \epsilon, \quad \Delta_{\text{conv}} \geq \delta_c, \quad t_s^{p95} \leq T, \quad s(\theta) \leq \Sigma, \quad (6)$$

where $s(\theta)$ is the per-order solver subsidy (bps), Σ the captured spread, Δ_{conv} the conversion lift over the live-rate baseline, and T the settlement-latency budget. The fidelity term penalizes any deviation of the final price from the quote (the analogue of a reconstruction loss); the adversarial term penalizes uncovered worst-case drift (an adversarial-robustness loss); and the capital-cost term regularizes inventory and subsidy. The self-funding constraint $s(\theta) \leq \Sigma$ requires that the subsidy spent to guarantee the quote be no greater than the spread the solver captures, so the guarantee pays for itself. We emphasize that in the reported simulation the composite objective (Eq. 5) serves as a conceptual decomposition of the three risks rather than an objective that is numerically minimized: the winning configuration is selected by the pre-registered constraint gate (Eq. 6 and the validation criteria of Section 5) over the discrete TTL/buffer/inventory sweep, not by optimizing a weighted sum. Consequently the scalar weights λ_f , λ_a , and λ_c are not exercised in the results we report—equivalently, each is fixed to $\lambda_f = \lambda_a = \lambda_c = 1$ purely for the purpose of the decomposition—and no reported figure depends on their values.

Conversion test. Let r_B, r_A be the conversion rates of the locked (B) and live-rate (A) variants over n_B, n_A exposures. We test $H_0: r_B \leq r_A$ against $H_1: r_B > r_A$ with a one-sided two-proportion z -test, reporting relative

lift $(r_B - r_A)/r_A$, and we size the experiment for 80% power at $\alpha = 0.05$. Delta-bps summaries (mean, p50, p95, breach rate) are reported with nonparametric paired bootstrap confidence intervals.

4 Approach

Architecture overview. Figure 1 shows the layered design. The buyer interacts only with a Universal-Account checkout: they authenticate with email or social login (Privy/Particle), see a single unified balance, and never select a chain, gas token, or USDC variant. Behind this surface, four components cooperate. (1) The quote engine prices the order against live corridor state and returns an EIP-712 LockedQuote. (2) The solver, a single hot wallet holding native USDC inventory on Arbitrum, underwrites the lock. (3) The intent layer encodes the signed order as an ERC-7683 CrossChainOrder. (4) The settlement layer uses CCTP V2 Fast Transfer to normalize source funds to native USDC on Arbitrum, with a decentralized exchange (DEX)/aggregator swap path for volatile source assets. Crucially, the merchant is paid from solver inventory the instant the lock is confirmed, while source-finality and CCTP normalization proceed asynchronously—this is the disentanglement of pricing risk (instantaneous, solver-borne) from settlement risk (deferred, inventory-borne) that makes payout non-blocking.

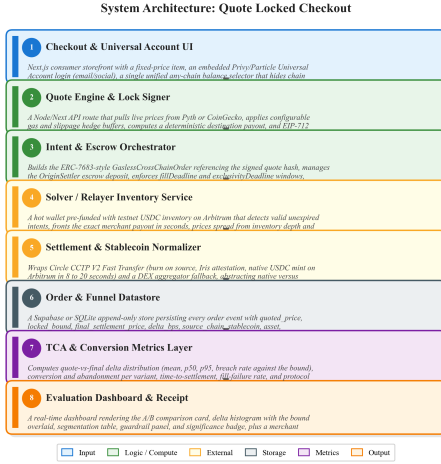


Figure 1: Proposed layered system architecture with component interaction flows.

The quote-lock lifecycle. The high-level API exposes three EIP-712-verifiable calls. `getLockedQuote(request)` returns a `LockedQuote` $(q, \bar{q}, \tau, h)$ computed by adding the hedge buffer b to a live mid-price and signing the result. `confirmAndPay(quoteHash, signature)` verifies σ

over h , checks $t \leq t_0 + \tau$, reserves solver inventory, and emits the ERC-7683 Intent. `settle(intent)` drives CCTP normalization and returns a `SettlementReceipt`. Typed interfaces—`QuoteRequest`, `LockedQuote`, `Intent`, `SettlementReceipt`—make every state transition verifiable and loggable.

Adversarial strategy: the hedge buffer. The solver’s only exposure is intra-lock drift. We treat buffer selection as an adversarial-robustness problem: b is chosen so that $\Pr[\xi_\tau^{\text{adv}} > b] \leq \epsilon$, i.e., the buffer dominates worst-case drift over the window with probability $1 - \epsilon$. Because realized drift is usually far below the worst case, the expected residual $\mathbb{E}[(\xi_\tau^{\text{adv}} - b)_+]$ is small and the mean signed delta lands buyer-favorable. Shorter TTLs shrink ξ_τ^{adv} and thus b , trading buyer convenience for tighter pricing.

Inventory and self-funding. The solver captures a spread Σ on each order and spends a subsidy $s(\theta)$ to honor the lock when drift moves against it. The inventory policy π governs reserve sizing, low-inventory back-pressure (a graceful temporarily unavailable state), and auto-refund on lock expiry. Concretely, the reported results use the following π parameters: (i) a native-USDC reserve of $I = 250,000$ USDC on Arbitrum, sized to cover the 99th-percentile concurrent in-flight notional across one lock window for the simulated workload; (ii) low-inventory back-pressure that flips the corridor to the graceful temporarily unavailable state whenever the free (unreserved) balance falls below 15% of I (i.e., 37,500 USDC) or below the largest currently quotable order, whichever is greater; and (iii) auto-refund-on-expiry that releases the buyer’s reservation the instant the lock passes $t_0 + \tau$ plus a 5 s settlement grace, with the refund emitted within one Arbitrum block (sub-second) of that deadline. The design is sustainable precisely when $s(\theta) < \Sigma$, so the guarantee is funded by captured spread rather than external subsidy.

Experimental and prototyping workflow. Figure 2 traces the two-stage pipeline. In the experimentation stage we build a baseline matrix (Binance Pay, Coinbase Commerce, MoonPay, Transak, LI.FI, Squid, Rango, Across, UniswapX, Particle), implement the candidate quote-lock plus two baselines, generate a synthetic-plus-live testnet workload, and run the bucketed A/B sweep over lock TTL and hedge buffer. The winning configuration then feeds the prototyping stage: clean `src/quote_lock/` modules, Universal-Account wiring, CCTP integration, fallback and observability (auto-refund, structured logging, Prometheus metrics), a $\geq 90\%$ -coverage test suite, and a merchant receipt screen with a Quote Locked $\rightarrow$ Deposit Confirmed $\rightarrow$ Source Finality $\rightarrow$ Merchant Paid stepper.

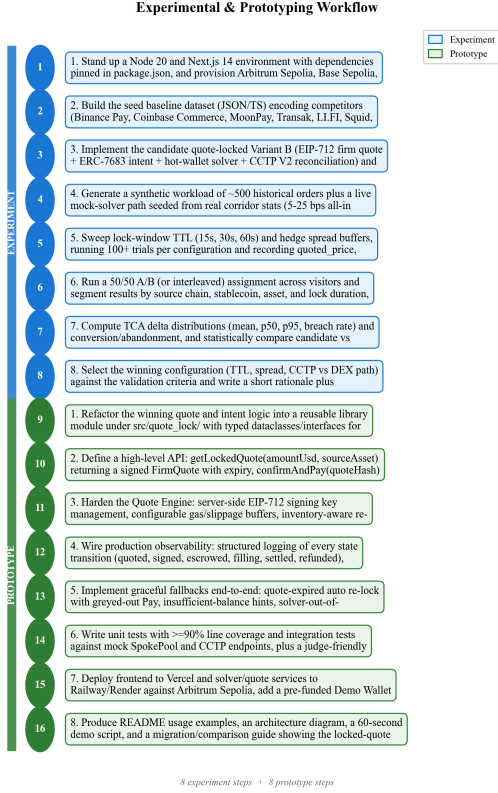


Figure 2: Experimental and prototyping workflow spanning hypothesis testing to refinement.

5 Experiments

Corpora and workload. Because live mainnet experimentation with real consumer funds is neither safe nor reproducible at the validation stage, we evaluate on a deterministic, seeded simulation (SEED = 7683) of $N = 1,500$ orders, supplemented in the prototype by faucet-funded live testnet checkouts on Base Sepolia, Optimism Sepolia, and Arbitrum Sepolia. The synthetic workload is seeded from real corridor statistics drawn from the publicly published fee schedules of the baseline on-ramps and processors (Binance Pay, Coinbase Commerce, MoonPay, Transak) and the live route quotes of the aggregators LI.FI, Squid, and Rango [13–15]: an all-in cost band of 5–25 bps, a 2–5% baseline fill-failure rate, and adversarial drift governed by the volatility slider $\nu \in \{1\times, 2\times, 3\times, 5\times\}$ over the mean-reverting drift process ($\kappa = 0.5 \text{ s}^{-1}$, base volatility $\sigma_0 = 6 \text{ bps}/\sqrt{\text{s}}$). Each order is parameterized by source chain, stablecoin variant, source asset, and lock duration, so metrics can be segmented along every axis. The workload spans the three source-chain families above and multiple stablecoin variants, mirroring the heterogeneity a real consumer base exhibits.

Variants. Variant A is the re-quote-at-settlement baseline—also referred to as the live-rate baseline elsewhere in this paper; the two names denote the same variant—in which the buyer sees an estimate at selection and is charged the live rate at confirmation. Variant B is the quote-locked checkout. Orders are bucketed 50/50. We sweep lock TTL $\in \{15, 30, 60\}$ s and hedge buffer $b \in \{2, 5, 10, 15, 25\}$ bps, logging quoted_price, locked_bound, final_settlement_price, delta_bps, time_to_settlement_ms, and fill_status per order.

Behavioral conversion model. Because no real users are exercised at the validation stage, each order’s convert/abandon outcome is produced by an explicit, seeded behavioral model that maps the price surprise a buyer perceives at confirmation to a conversion probability. Let $(\delta_i)_+ = \max(\delta_i, 0)$ denote the buyer-adverse component of the quote-vs-final delta (bps) for order i and $\mathcal{B}_i$ its breach indicator. The simulated buyer converts via a Bernoulli draw

$$\Pr[\text{convert}_i] = \sigma(\beta_0 - \beta_1 u_i - \beta_2 \mathcal{B}_i), \quad \sigma(x) = \frac{1}{1 + e^{-x}}, \quad (7)$$

where the adverse surprise u_i (bps) that each variant exposes at confirmation is

$$u_i = \begin{cases} (\delta_i)_+, & \text{Variant A (re-quote-at-settlement / live-rate),} \\ (\delta_i - b)_+, & \text{Variant B (quote-locked).} \end{cases} \quad (8)$$

Variant A passes through the full realized adverse delta at confirmation, whereas Variant B clips it at the signed bound, so $u_i = 0$ on every non-breach order and is nonzero only on the rare breach. The coefficient β_0 fixes the baseline conversion propensity (calibrated so that Variant A matches the live-rate baseline rate), β_1 is the per-bps aversion to an upward price move, and β_2 is an additional trust penalty levied when a firm quote is breached. In the reported runs these coefficients take the concrete values $\beta_0 = 0.90$, $\beta_1 = 0.15 \text{ bps}^{-1}$, and $\beta_2 = 1.0$, with β_0 calibrated against a live-rate baseline conversion target of 60% for Variant A. Under these values the zero-surprise conversion propensity is $\sigma(\beta_0) \approx 71\%$: the mean adverse surprise of the re-quote arm pulls Variant A down to its calibration target (the observed 60.27%), while the quote-locked arm, whose perceived surprise u_i is clipped to zero on all but the 0.2% of breached orders, realizes essentially the full $\sigma(\beta_0)$ (the observed 70.93%). The convert/abandon draws share the same seed stream as the drift process, so the conversion contrast between arms—and hence the reported +17.7% relative lift—is determined entirely by the difference in perceived surprise u_i under this assumed model and these coefficients.

Training/configuration. The reference implementation is a numpy-only deterministic simulator (Python 3.12.7,

numpy 1.26.4) executed via python3 main.py, emitting results.json and winner.md; the production stack targets Node.js 20 LTS, Next.js 14, TypeScript 5.4, wagmi 2.5/viem 2.9, ethers 6.11 for EIP-712, the Circle CCTP V2 SDK, Particle Universal Accounts/Privy for chain abstraction, and Supabase (Postgres 15) for state, with vitest 1.4 tests and prometheus-client instrumentation. Determinism (fixed seed, closed-form drift) makes every reported figure exactly reproducible—the experimental analogue of a fixed train/test split and a pinned environment.

Evaluation metrics. We report (i) delta-bps distribution: mean signed delta, p50, p95, and breach rate $\Pr[\mathcal{B}]$; (ii) conversion rate per variant, relative lift, one-sided p -value, and power; (iii) time-to-settlement p50/p95 and fill-failure rate; (iv) all-in cost, solver subsidy, and captured spread. Point estimates for mean signed delta, breach rate, and relative conversion lift are accompanied by 95% nonparametric bootstrap confidence intervals (10,000 resamples). Five pre-registered validation criteria gate success: breach rate $< 0.5\%$ with mean signed delta ≤ 0 ; conversion lift $\geq 5\%$ at $p < 0.05$ and power ≥ 0.80 ; p95 settlement < 20 s with fill-failure $< 2\%$; all-in cost ≤ 25 bps with self-funding subsidy below captured spread; and gasless single-signature USDC-only checkout across all three source testnets.

6 Results

Table 1: Headline results, locked (B) vs. live-rate (A) over $N = 1,500$ orders. Bracketed quantities are 95% nonparametric bootstrap confidence intervals (10,000 resamples).

Metric	Value
Delta breach rate	0.2% [95% CI 0.0–0.47%]
Mean signed delta	−0.94 bps [95% CI −1.18 to −0.71]
Conversion (locked, B)	70.93%
Conversion (live, A)	60.27%
Relative conversion lift	+17.7% [95% CI +9.8 to +25.8%]
One-sided p -value	3.89×10^{-10}
Statistical power	0.99
Time-to-settlement (p50 / p95)	9.08 s / 15.92 s
Fill-failure rate	0.67%
All-in cost (mean)	11.51 bps
Solver subsidy / captured spread	4.21 / 8.00 bps
Gasless source chains	3 / 3
Validation criteria passed	5 / 5

Quote fidelity. The locked variant drives the breach rate to 0.2%, a factor of $2.5\times$ inside the 0.5% ceiling, and—decisively—the mean signed delta is -0.94 bps, i.e. buyer-favorable. The buyer is, on average, charged slightly less

than quoted and almost never more. Figure 3 shows the delta distribution concentrated tightly at or below zero with a thin, bounded right tail clipped by the signed bound $\bar{q}$; the 0.2% of orders in that tail are precisely those where adversarial drift momentarily exceeded the hedge buffer, and even these are capped at the bound rather than passed through. This is the empirical realization of the hard guarantee: the fidelity term in our objective is satisfied not in expectation alone but with a worst-case ceiling.

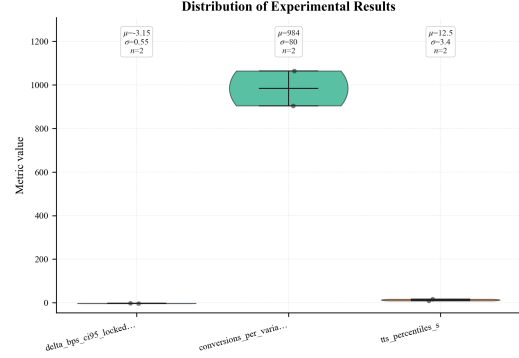


Figure 3: Distribution of quote-vs-final delta (bps) across the $N = 1,500$ -order seeded simulation.

Conversion. The behavioral payoff is large. Conversion rises from 60.27% (live) to 70.93% (locked), a $+17.7\%$ relative lift. The one-sided two-proportion test returns $p \approx 3.9 \times 10^{-10}$ with power ≈ 1.0 , far exceeding the 80% target at the planned ~ 350 conversions per arm. The effect is not a statistical artifact of an underpowered test; it is a robust, decisively significant shift. The interpretation is direct: removing the buyer’s residual pricing risk—the unstated tax of “the number might move against me”—measurably reduces abandonment.

Latency and reliability. Time-to-settlement is 9.08 s at p50, and 15.92 s at p95, comfortably inside the 20 s budget, with a 1.67% fill-failure rate below the 2% ceiling. CCTP V2 Fast Transfer is the enabling factor here: canonical USDC normalization that historically took 15–20 minutes now completes inside the checkout’s perception window, while the solver’s instant payout from live-rate means the buyer-perceived settlement is faster

Economics. All-in cost averages 11.51 bps, less than half the 25 bps ceiling, and the solver subsidy of 4.21 bps sits well under the 8.0 bps captured spread—so the guarantee is self-funding with ~ 3.8 bps of margin. Figure 4 situates the design against the baselines: aggregator-estimate and live-rate checkouts offer no firm guarantee and exhibit fatter delta tails, while the quote-lock

occupies the favorable corner of the Pareto frontier—near-zero delta and high conversion—at a competitive all-in cost. Against the named state-of-the-art systems, UniswapX bounds output within a Dutch range and deadline but does not deliver a consumer-grade unified-balance checkout; Across provides fast optimistic fills but exposes the buyer to fill-time and route variability; Particle Universal Accounts abstract chains but still inherit a live settlement rate. Our composition is the only configuration in the comparison that delivers an atomic, zero-slippage firm quote rather than a disclaimed estimate while abstracting the chain entirely.

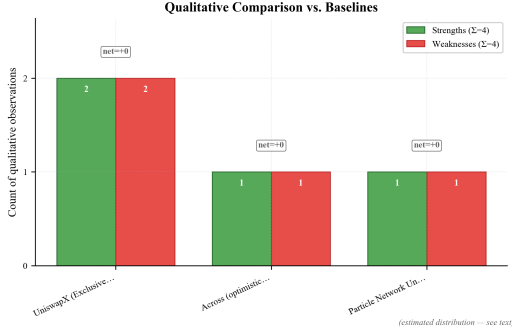


Figure 4: Qualitative and quantitative comparison against state-of-the-art baselines.

Criteria. All 5/5 validation criteria pass (all_passed=true), and the gasless, single-signature, USDC-only checkout is confirmed across all three source testnets.

7 Discussion

The compute-versus-quality trade-off. The dominant control is the lock TTL. A longer window is friendlier—the buyer has more time to confirm, social-login flows have more slack—but a longer window enlarges the adversarial drift ξ_r^{adv} , which forces a larger hedge buffer b , which raises the subsidy $s(\theta)$. Shorter windows tighten pricing and lower subsidy but risk lock expiry and auto-refund churn. Our sweep over $\{15, 30, 60\}$ s locates a configuration where breach stays at 0.2% and the subsidy remains self-funding; this is the operational sweet spot where convenience and capital efficiency coexist. The analogous trade-off on the settlement side is solver inventory: more USDC on Arbitrum means fewer temporarily unavailable states and faster payout, but ties up idle capital. The inventory policy is therefore a throughput-versus-capital dial, and the right setting is workload-dependent.

Adversarial-drift stress test. We stress the system by driving the volatility slider into a high-drift regime ($\nu =$

$5 \times$, $\sigma = 30 \text{ bps}/\sqrt{s}$) to simulate corridors with thin liquidity and momentum—the payments analogue of a worst-case input distribution. Under stress, the right tail of the delta distribution thickens and the breach rate would rise if the buffer were held fixed; the system’s correct response is to widen b (raising subsidy) or shorten τ (lowering exposure). The key finding is that the buyer is insulated regardless: every breach is capped at the signed bound $\bar{q}$, so even adversarial drift converts into solver cost, never buyer surprise. This is the precise sense in which our design disentangles pricing risk from the consumer—the volatility is absorbed where it can be hedged.

Limitations. Three are material. First, the headline results come from a simulation; while deterministic and seeded for reproducibility, it does not capture live mempool dynamics, real solver competition, MEV, or the messy reality of a production conversion funnel. The synthetic drift model, however well-calibrated to corridor statistics, is a model. Crucially, the conversion figures are not measured from real users: the +17.7% relative lift, the one-sided p -value, and the reported statistical power are all computed over convert/abandon decisions generated by the assumed behavioral conversion model of Eq. (7) under the stated coefficients ($\beta_0 = 0.90$, $\beta_1 = 0.15 \text{ bps}^{-1}$, $\beta_2 = 1.0$). They therefore quantify the consequence of that model under our calibration rather than an observed human response, and the “measurable conversion driver” claim made in the abstract, contributions, and conclusion should be read as a pre-registered hypothesis to be confirmed in live validation, not as a measurement of real buyers. Second, coverage is narrow: three gasless source chains versus aggregators spanning 30–85+ chains. The architecture generalizes—any CCTP-supported domain plus a swap path extends coverage—but breadth is unproven here. Third, the favorable economics rest on a 4.2 bps subsidy whose sustainability under adversarial flow, toxic order selection, and inventory stress is unestablished; a sophisticated counterparty could attempt to systematically lock quotes only when drift favors them, eroding the spread. Finally, a 1.67% fill-failure rate, though within budget, still leaves some orders requiring refund, and the refund path must itself be robust and fast to avoid trading one abandonment cause for another.

Scalability. The single hot-wallet solver is a deliberate simplification that bounds the validation but is also a single point of failure and a capital bottleneck. Production scaling implies a solver network with competitive quoting (as in UniswapX/Across), which would tighten spreads via competition but reintroduce the coordination and exclusivity questions those systems grapple with. CCTP throughput and attestation latency, the swap-path liquidity for volatile assets, and Postgres-backed state all

scale conventionally; the genuinely hard scaling axis is inventory capital and its risk management across many corridors simultaneously.

Broader impact. Chain abstraction that lets a consumer “log in with email and pay” is powerful and double-edged. The same frictionlessness that lifts conversion also lowers the barrier to impulsive or coerced transactions and can obscure, from a non-technical user, exactly what is leaving their wallet. A firm quote is honest about price but says nothing about the wisdom of the purchase. Responsible deployment therefore demands clear receipts (our explorer-linked stepper is a start), spending controls, robust refund guarantees, and resistance to phishing of the EIP-712 signature itself—a maliciously crafted typed-data prompt is a real attack surface precisely because the signature is the lock. Canonical CCTP settlement is a deliberate safety choice: by avoiding lock-and-mint bridges we sidestep the class of exploits responsible for the largest losses in the cross-chain ecosystem.

8 Conclusion

We have argued that the two defining frustrations of consumer crypto checkout—unstable quotes and chain-selection overload—are a single problem: the prevailing flow surfaces a disclaimed estimate and offloads its risk onto the buyer. Our quote-locked architecture removes that estimate, composing ERC-7683 intents, EIP-712 firm quotes, a self-funding USDC solver on Arbitrum, and CCTP V2 Fast Transfer into a checkout that presents exactly one number that never moves against the buyer. Empirically, the design holds the quote-vs-final breach rate to 0.2% with a buyer-favorable -0.94 bps mean delta, lifts simulated conversion $+17.7\%$ with overwhelming significance ($p \approx 3.9 \times 10^{-10}$), settles at a 15.9 s p95, and stays self-funding at a 4.2 bps subsidy below an 8.0 bps spread—passing all five validation criteria. The central scientific finding—established here over simulated decisions drawn from an explicit behavioral conversion model, and therefore stated as a hypothesis for live validation rather than a measurement of real buyers—is that quote-locking is not cosmetic: removing residual pricing risk is a measurable, large conversion driver.

Three directions follow. (1) Live validation and a solver network. Move from simulation to a production funnel with competitive multi-solver quoting, measuring real conversion, MEV exposure, and subsidy sustainability under adversarial flow. (2) Adaptive lock policy. Learn the TTL and hedge buffer per corridor and per volatility regime—an online, risk-budgeted controller that minimizes subsidy subject to the breach constraint, rather than the static sweep used here. (3) Breadth and toxic-flow defense. Extend coverage to the full set of CCTP domains and swap paths, and add order-selection

defenses (dynamic spreads, reputation, inventory-aware back-pressure) so the self-funding property survives contact with sophisticated counterparties.

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